Mini-project — Gas liquefaction (LNG)

Design and comparison of liquefaction cycles

1. Introduction

Gas liquefaction turns a gas into its liquid by cooling it below its boiling point, so that a large amount of energy can be stored or transported in a small volume. Liquefied natural gas (**LNG**) is methane cooled to about $-162\,^{\circ}\mathrm{C}$ at atmospheric pressure; liquefying it shrinks its volume roughly 600-fold, which is what makes long-distance transport and large-scale storage of natural gas possible.

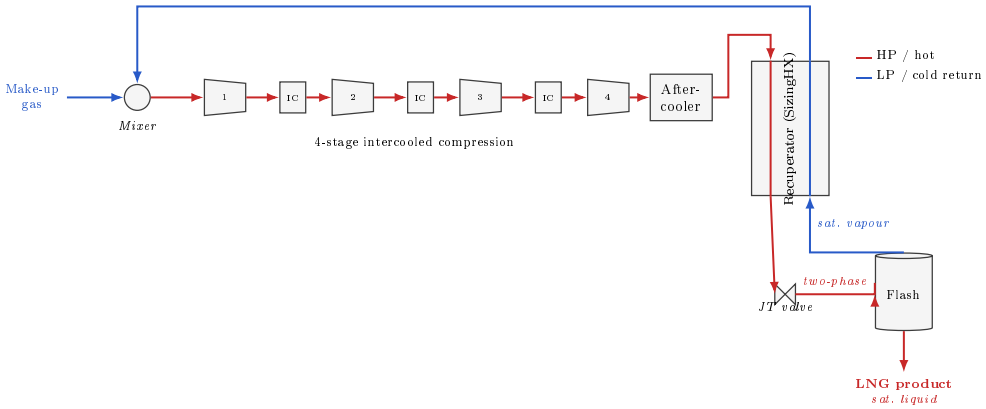
Cooling a gas down to cryogenic temperatures is energy-intensive and relies on two effects: the **Joule–Thomson effect** (a real gas cools when throttled from high to low pressure) and **recuperation** (the cold, un-liquefied vapour is used to pre-cool the incoming high-pressure gas). The simplest industrial cycle, **Linde–Hampson**, combines compression, a recuperative heat exchanger, a throttling valve and a flash separator. Its liquefied fraction per pass is modest, so several classic improvements exist. This project asks each group to quantify what one such improvement brings compared with the basic cycle.

2. Context

Each group studies **two systems**: a common **baseline cycle L0** (the simple Linde–Hampson cycle, studied by everyone) and **one advanced cycle L1–L3** (one per group). The goal is a clear **base-vs-advanced comparison**; at the end, the groups pool their results and compare the advanced cycles against the shared baseline.

All systems share the same case study and the same fixed parameters (Section 3), so the comparison is meaningful. The work is in **design mode only**: all operating conditions are fixed, and the cycle is solved at a single steady-state operating point. A *parametric study* simply means re-solving this design at each value of the swept parameter.

3. The baseline cycle — L0 (Linde–Hampson)



The make-up gas is mixed with the recycled vapour, compressed to P_{high} and cooled back to $T_0 + 5$ K. The high-pressure gas is then cooled further in the **recuperative heat exchanger** by the returning cold vapour, throttled through the **Joule–Thomson valve** to 1 bar (which produces

a two-phase mixture), and sent to the **flash separator**. The saturated liquid is withdrawn as product; the saturated vapour returns through the recuperator (warming up as it pre-cools the incoming gas) and is recycled to the compressor.

L0 is built by every group and serves as the reference against which the advanced cycle is compared. The specification and assumptions below apply to L0 and to all advanced cycles.

Working fluid and reference conditions

Quantity	**Value**	**Note**
Gas	Methane	proxy for natural gas (assumed); real-gas properties (CoolProp)
Make-up (feed) gas	25 °C, 1 bar	replaces the liquefied product
Liquid product	saturated liquid at 1 bar	drawn from the flash separator ($\approx -162\,^{\circ}\mathrm{C}$)
Ambient (heat rejection)	$T_0 = 25$ °C	reference for the coolers (coolant side)

Fixed parameters

Parameter	Value	Note
High pressure P_{high}	100 bar	default; one of the parameters to optimise
Compression	multi-stage, intercooled to $T_0 + 5$ K	equal pressure ratios; 4 stages for all cycles (see below)
Compressor isentropic efficiency	0.75	per stage
Cooler pinch (inter/after-coolers)	5 K	gas cooled to $T_0 + 5$ K $= 30\,^{\circ}$C
Recuperative-HX pinch	5 K	minimum approach, imposed at one terminal set by the capacity rates (see assumption C); internal pinch checked a posteriori
Recuperator film coeff. (hot, HP side)	1000 W/m^{2}K	dense high-pressure gas
Recuperator film coeff. (cold, LP side)	500 W/m^{2}K	low-pressure return vapour; area ratio 1:1
Joule–Thomson valve	isenthalpic to 1 bar	
Flash separator	at 1 bar	saturated liquid (product) + saturated vapour (recycled)
Liquid product flow $\dot{m}_{\text{liq}}$	1 kg/s	sets the plant scale

Compression harmonised to 4 stages for all cycles. To keep the model identical across groups, compression is always $1 \to 3.16 \to 10 \to 31.6 \to 100$ bar ($r \approx 3.16$ per stage). This places a stage boundary exactly at $P_{\text{int}} = 10$ bar, which L2 needs for the MP re-injection; L0, L1 and L3 use the same 4-stage split so that the compression model is common and the comparison is clean.

Only the **recuperator** is sized as an area exchanger (`SizingHX`); the inter/after-coolers and the L1 precooler are modelled as simple thermal duties (`SimpleHeater`, no area), as a simplification. The film coefficients above are what turn the recuperator duty into a comparable area.

Modelling assumptions

A. Design mode only — no simulation mode (all conditions fixed). Steady-state resolution (`solve`).

B. Real-gas properties (CoolProp) are mandatory: the Joule–Thomson cooling vanishes for an ideal gas.

C. Heat exchangers: single-zone `SizingHX` with constant c_p, used as-is including for the cryogenic recuperator (large temperature span) and any two-phase stream. This is an assumed

approximation and must be stated explicitly in the code/comments. Being single-zone, the 5 K pinch is imposed at ONE terminal. **Terminology:** in the counter-flow recuperator the **warm end** is the physical end where the hot stream *enters* and the cold stream *leaves* (both streams are hot there); the **cold end** is where the hot stream *leaves* and the cold stream *enters* (both are cold). Note that the hot stream's *outlet* is at the cold end, not the warm end. **Which terminal carries the pinch is not a free choice: it is fixed by the heat-capacity rates.** Before fixing the pinch side, compute $\dot{m}\,c_p$ of each stream: the minimum approach sits at the **warm end** if $(\dot{m}c_p)_{\text{hot}} > (\dot{m}c_p)_{\text{cold}}$, and at the **cold end** if $(\dot{m}c_p)_{\text{hot}} < (\dot{m}c_p)_{\text{cold}}$ (the stream with the smaller $\dot{m}c_p$ has the larger temperature swing and closes the gap). Imposing the pinch at the wrong terminal produces a temperature crossover and an infeasible design. This gives the **warm end** for L0, L1 and L2 (the HP hot stream carries the full circulating flow and a higher c_p, so $(\dot{m}c_p)_{\text{hot}} > (\dot{m}c_p)_{\text{cold}}$). For L3, the pinch position depends on the fraction of the flow going through the expander, so compute $\dot{m}c_p$ for the chosen x_e to decide which terminal applies. The true minimum approach is internal and must still be **checked a posteriori** (the mono-zone model can be optimistic near methane's pseudo-critical region; in practice this check matters most for L1 and L3, where the two T–h curves run closest).

D. Compression: modelled as multi-stage with intercooling back to $T_0 + 5$ K between stages (equal pressure ratios); the compression work is the sum over stages. Fixed at 4 stages at 100 bar for all cycles (per-stage ratio $r \approx 3.16$).

E. Components to reuse: `FlashTank` (saturated-liquid / saturated-vapour separator), `Mixer` (make-up gas + recycled vapour), `Valve` (isenthalpic throttle, outlet pressure set by a system constraint), and `Splitter` (L3, to divert the expander fraction). The expander (L3) reuses the turbine model with an isentropic efficiency.

F. Steady-state mass balance: the make-up gas equals the liquid product; the un-liquefied vapour is recycled. The liquefied fraction per pass sets the ratio between the circulating flow and the product.

Starting the recycle loop

Every cycle has a **recycle loop**: the un-liquefied vapour returns to the compressor, so the circulating flow is not known until the liquefied fraction y is solved. This is a fixed point and needs an initial guess. Start from $y_0 \approx 0.1$ (circulating flow $\approx 10\,\dot{m}_{\text{liq}}$) and let `solve` converge on the loop. As a sanity check, **L0 should give** $y \approx 9\%$ at the default point; a value far from this (e.g. 50% or a few %) signals a sign, unit, or pinch-terminal error rather than a converged solution.

Comparison metrics

At the fixed design point, each group reports and compares against L0:

- **Liquefaction yield** $y = \dot{m}_{\text{liq}}/\dot{m}_{\text{circulating}}$ (fraction liquefied per pass);
- **Specific liquefaction work** $w = W_{\text{total}}/\dot{m}_{\text{liq}}$ (kJ per kg of LNG) — the headline figure. W_{total} is the compressor work; for L1 add the precooling work $W_{\text{pre}} = Q_{\text{pre}}/\text{COP}_{\text{pre}}$; for L3 it is **net** of the recovered expander work, $W_{\text{total}} = W_{\text{comp}} - W_{\text{exp}}$. State the convention so the headline figures stay comparable;
- **Heat-exchanger sizes** (areas, from the design solve) — only the recuperator(s) carry an area;
- optionally, a **figure of merit** $w_{\min}/w$ against the ideal minimum liquefaction work. The ideal minimum is the reversible (exergetic) liquefaction work from the feed state (25 °C, 1 bar) to the product (saturated liquid, 1 bar), rejecting heat at T_0:

$$w_{\min} = T_0\,(s_{\text{feed}} - s_{\text{liq}}) - (h_{\text{feed}} - h_{\text{liq}}),$$

evaluated with CoolProp. For methane this is ≈ 1083 kJ/kg, so L0 has a figure of merit of $\approx 9\%$.

Expected L0 results (order of magnitude)

To help each group check its baseline before building the advanced cycle, L0 at the default point ($P_{\text{high}} = 100$ bar, 4 stages) should land near:

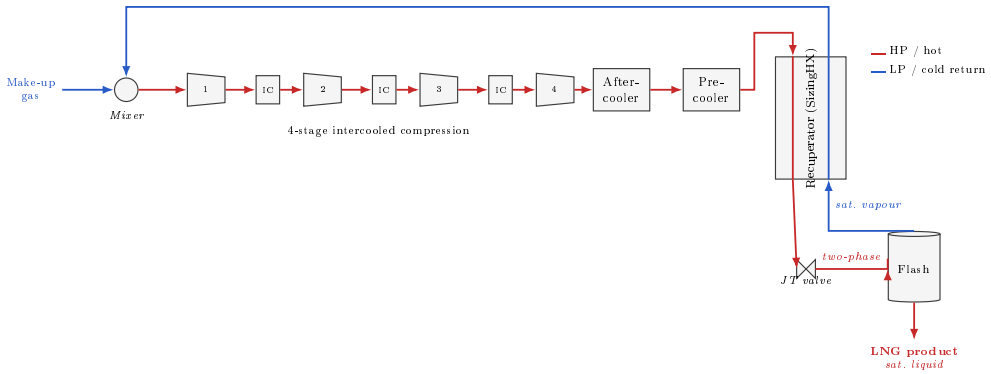
Quantity	Order of magnitude	Note
Liquefaction yield y	$\approx 9\%$	fraction liquefied per pass
Circulating flow	≈ 11 kg/s	for $\dot{m}_{\text{liq}} = 1$ kg/s
Specific work w	≈ 1.1–1.2×10^4 kJ/kg	≈ 3–3.4 kWh per kg LNG
Figure of merit $w_{\min}/w$	$\approx 9\%$	$w_{\min} \approx 1083$ kJ/kg

These are indicative, not targets to reproduce exactly (they shift with the number of stages and P_{high}); they are there to catch gross errors.

4. Advanced cycles

Each group develops one of the following, in addition to L0. The common specification (Section 3) applies throughout; only the points below are specific to each cycle. The three subsections are independent of one another.

L1 — Precooled Linde–Hampson

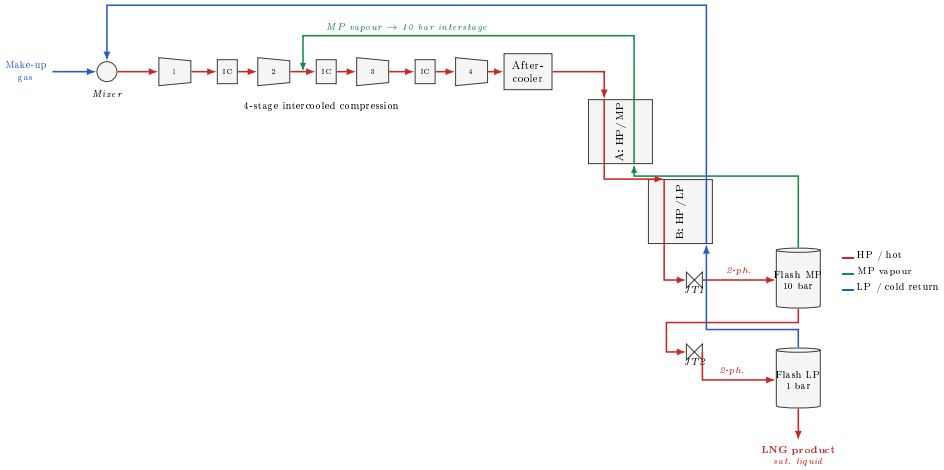


A **precooler** cools the high-pressure gas to a fixed temperature $T_{\text{pre}} = -35\,^{\circ}\text{C}$ before the recuperator, using an auxiliary refrigeration loop. The precooling duty is not free: its work is charged through an assumed refrigeration coefficient of performance $\text{COP}_{\text{pre}} = 2$, i.e. $W_{\text{pre}} = Q_{\text{pre}}/\text{COP}_{\text{pre}}$, and added to W_{total}. Expected effect: a strong increase in the liquefaction yield.

The L1 group is required to carry out two extra analyses:

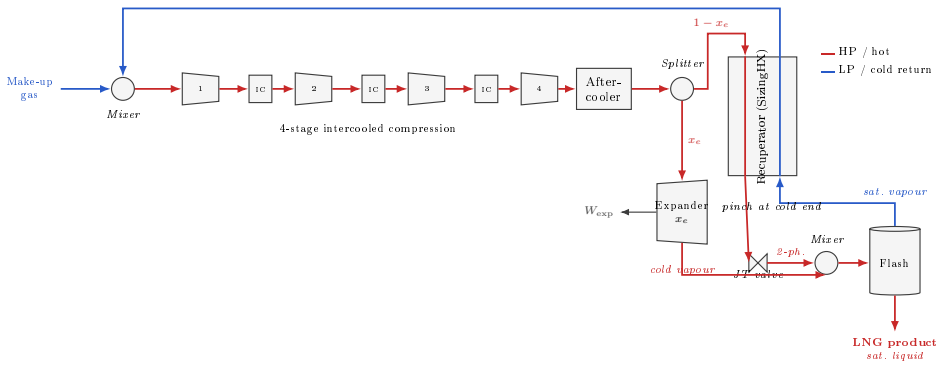
1. **Effect of the number of compression stages**: vary the stage count (e.g. 1 to 5, equal ratios) and plot the specific work and yield versus the number of stages; comment on the trade-off (compression work vs added complexity).
2. **Exergy analysis**: build the exergy balance of the cycle, locate the main irreversibilities (compression, recuperator, throttle, precooler) and report the exergetic efficiency. Compare the exergy destruction of L1 against L0.

L2 — Dual-pressure Linde cycle



The expansion is split over **two pressure levels**: the high-pressure liquid is throttled first to an intermediate pressure P_{int} (JT 1) and flashed; the **medium-pressure vapour** is returned to the compressor interstage (reducing the compression work), while the **liquid** is throttled again (JT 2) to 1 bar and flashed, giving the product. The low-pressure vapour returns through the recuperator. Default $P_{\text{int}} = \sqrt{P_{\text{low}}\, P_{\text{high}}} = 10$ bar, one of the parameters to optimise. Reuses the `FlashTank` twice. Because the MP vapour must re-enter the compressor **at its own pressure**, the compression has a **stage boundary exactly at $P_{\text{int}} = 10$ bar** — which the harmonised 4-stage split ($1 \to 3.16 \to 10 \to 31.6 \to 100$ bar) provides, the MP vapour re-injected at the 10 bar node. The MP vapour is re-injected **at its own temperature** — it is neither heated nor cooled before mixing. It is cold (it leaves the recuperator cold), which *lowers the inlet temperature of the upper compression stages and therefore reduces the compression work*: this is part of the benefit of the dual-pressure arrangement. The recuperator is **split into two sections in series**: one exchanging the HP stream against the MP vapour, the other against the LP vapour. Expected effect: lower compression work.

L3 — Claude cycle (with expander)



A fraction x_e of the high-pressure gas (diverted at 30 °C, *before* the recuperator) is sent to a **turbo-expander** instead of being throttled: the expander produces work and cools the gas more effectively than a valve; its cold outlet is mixed with the throttled stream (a `Mixer`) just before the flash separator. The remaining fraction follows the usual throttle-and-flash path. Default expander fraction $x_e = 0.5$ (parameter to optimise) and expander isentropic efficiency 0.80. The recovered expander work is subtracted from the total: $W_{\text{total}} = W_{\text{comp}} - W_{\text{exp}}$. With the inlet at

30 °C, the expander outlet at 1 bar is single-phase vapour (≈ -153 °C, a few K of superheat), so the turbine stays dry. Expected effect: better cooling and higher yield, with work recovered at the expander.

Numerical note — warm-start on x_e. Solving L3 directly at $x_e = 0.5$ may fail to converge. Use a **continuation**: start at a small x_e (close to L0, whose solution is known), then increase x_e step by step, each solve warm-started from the previous solution.

```
def warm_start(sys_):
    for v in sys_.unknowns:
        v.start = v.value # the current solution becomes the starting point

# after a successful solve at x=0.1 (seeds + autostart already done):
for xe in [0.01, 0.05, 0.1, 0.15, 0.2, 0.3, 0.4, 0.5]:
    warm_start(sys_) # restart from the previous solution
    x_expander = xe # the split-constraint lambda reads this global
    sol = solve(sys_, steps=40)
    print(f"x_e={xe:.2f} success={sol.success} m_circ={mixer.out.mdot:.2f}")
    if not sol.success:
        break
```

5. Expected deliverables

This is an **open project**: beyond the fixed design point, each group is free to propose analyses, parameter changes or circuit modifications that improve performance. Each group submits **two deliverables**:

1. **A project report** (max. 10 pages) presenting the model and an analysis of the results: liquefaction yield, specific work, the split between compression (and precooling for L1) and any recovered work, heat-exchanger sizes, and energy/exergy balances — with a clear **base-vs-advanced comparison**.
2. **Python code**: commented, in a **single file** (notebook or `.py`) containing the model and the analyses presented in the report. The model is built on the shared library and reuses the components listed in Section 3 (`FlashTank`, `Mixer`, `Valve`, `Splitter`, expander where required).

Examples of extensions that exist in this problem (useful for the open-ended work) include: parametric studies of P_{high}, of the intermediate pressure P_{int} (L2), of the expander fraction x_e (L3), of the precooling temperature (L1) or of the heat-exchanger pinch; combining precooling with an expander; multi-stage expansion; or alternative gases and feed conditions. The quality of the reasoning and of the quantified comparison matters more than finding a single "best" answer.